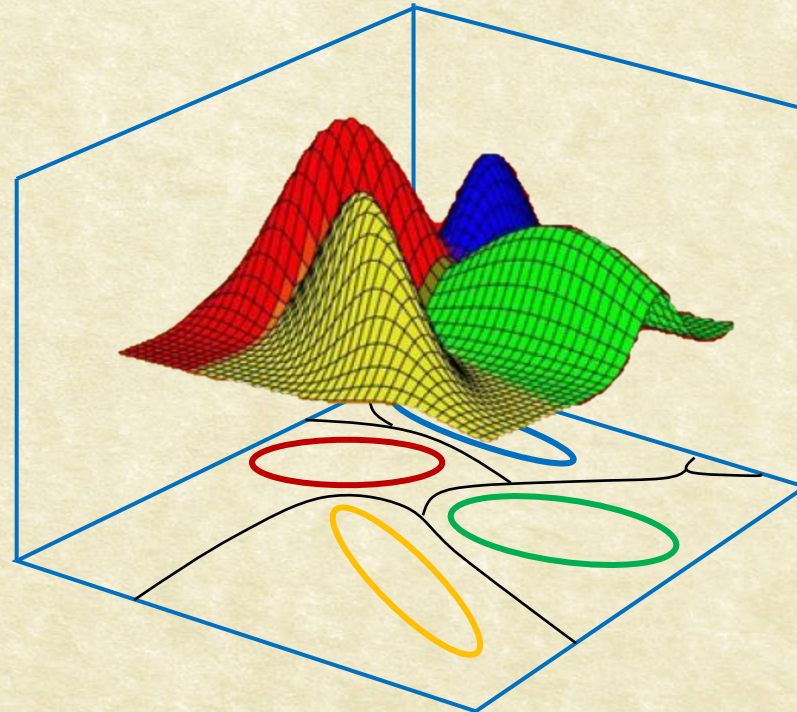




CS7.404: Digital Image Processing

Monsoon 2025: Fourier Transform - 2

A Closer Look



Raghavendra G S

CVIT,

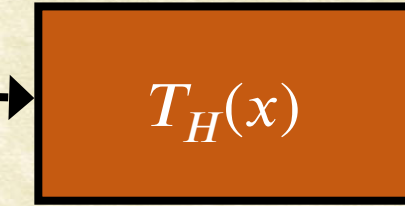
IIIT Hyderabad



Processing in Spatial vs Frequency Domain

Spatial/Temporal
Domain

$f(x)$



$g(x)$

Transform

$\mathcal{F}[f(t)]$

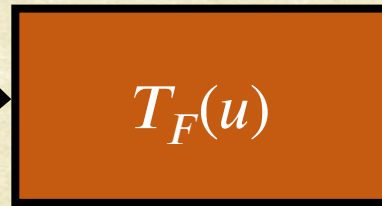


Inverse Transform

$\mathcal{F}^{-1}[F(\omega)]$

Frequency Domain

$F(u)$



$G(u)$



Fourier Transform (FT): Recap

- The Fourier Transform of a function $f(t)$ is defined by:

$$F(\omega) = \mathcal{F}[f(t)] = \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt$$

- The Inverse Fourier Transform (IFT) is given by:

$$f(t) = \mathcal{F}^{-1}[F(\omega)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega)e^{i\omega t} d\omega$$



Fourier Series: Recap

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{i\frac{2\pi n}{T}t}$$

$$c_n = \frac{1}{T} \int_0^T f(t) e^{-i\frac{2\pi n}{T}t} dt$$

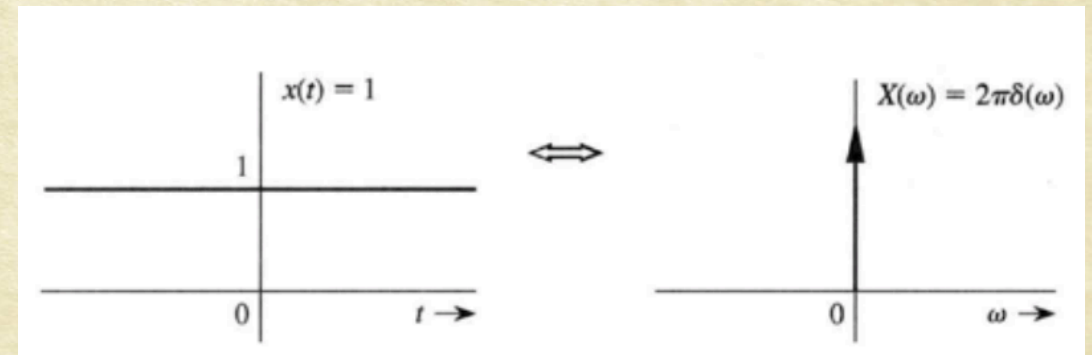
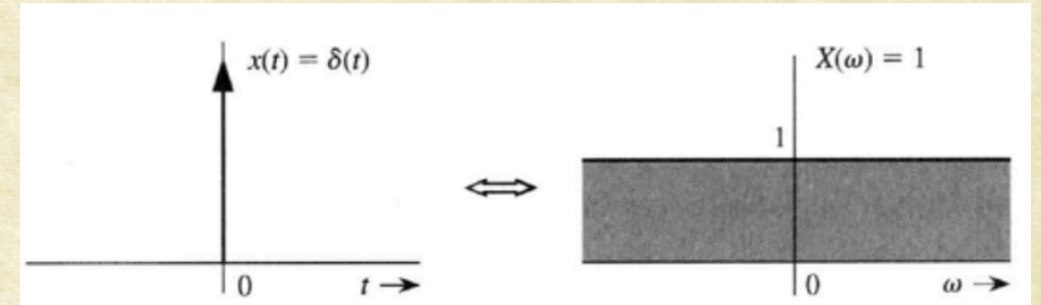
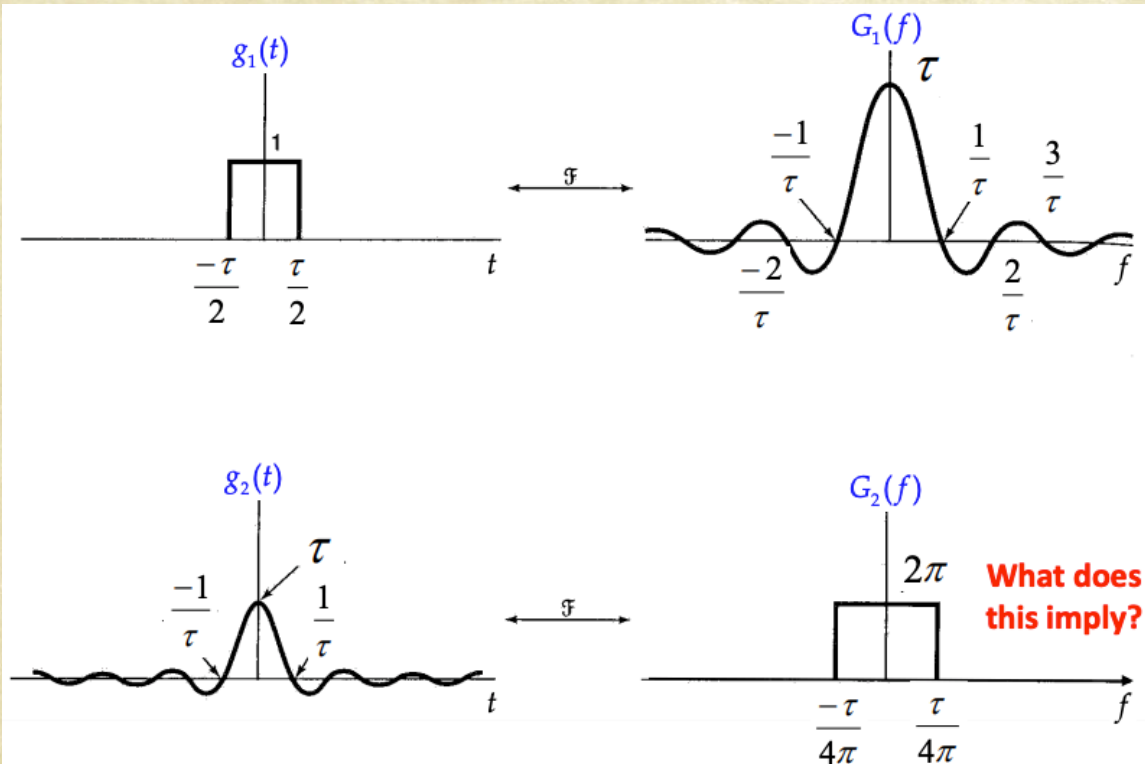


Symmetry property of FT

$$\mathcal{F}[f(t)] = F(\omega)$$

$$\implies \mathcal{F}[F(t)] = 2\pi f(-\omega)$$

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt$$
$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega)e^{i\omega t} d\omega$$





Fourier Transform of a Square Pulse

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt; \quad F(\mu) = \int_{-\frac{W}{2}}^{\frac{W}{2}} Ae^{-i2\pi\mu t} dt$$

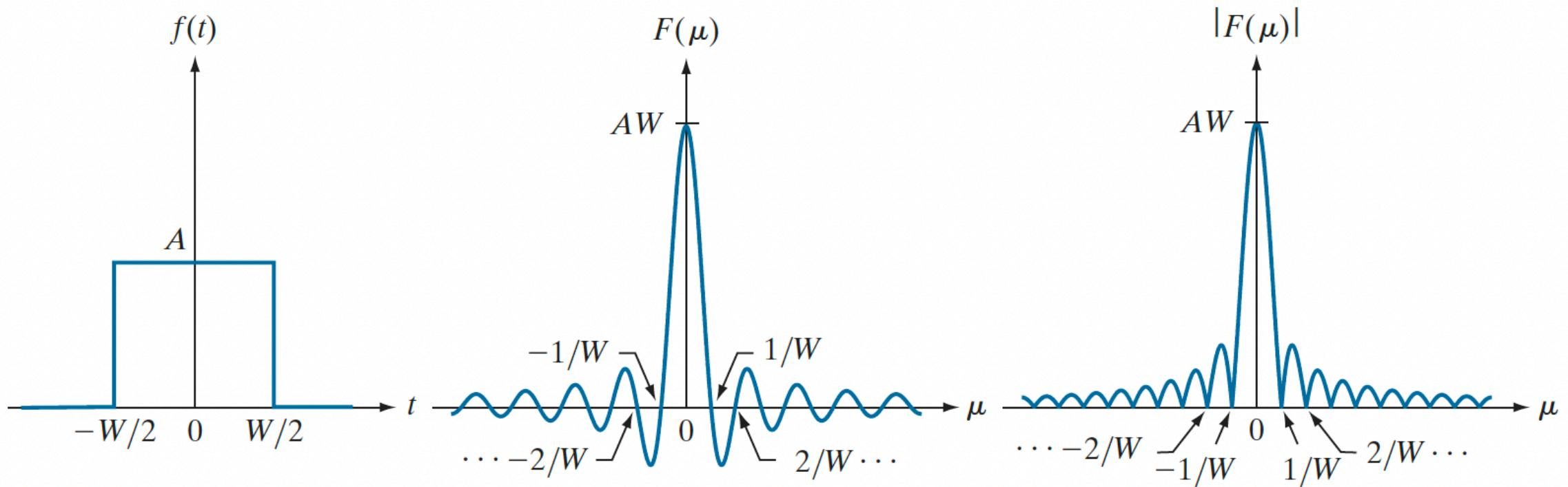


Fig: 4.4; Gonzalez and Woods



Fourier Transform of a Square Pulse

$$f(\mu) = \int_{-\frac{W}{2}}^{\frac{W}{2}} A e^{-i2\pi\mu t} dt$$



Fourier Transform of a Square Pulse

$$f(\mu) = \int_{-\frac{W}{2}}^{\frac{W}{2}} A e^{-i2\pi\mu t} dt$$

$$= \frac{-A}{i2\pi\mu} \left[e^{-i2\pi\mu t} \right]_{-W/2}^{W/2}$$



Fourier Transform of a Square Pulse

$$F(\mu) = \int_{-\frac{W}{2}}^{\frac{W}{2}} A e^{-i2\pi\mu t} dt$$
$$= \frac{-A}{i2\pi\mu} \left[e^{-i2\pi\mu t} \right]_{-W/2}^{W/2}$$

$$= \frac{-A}{i2\pi\mu} \left[e^{-i\pi\mu W} - e^{i\pi\mu W} \right]$$



Fourier Transform of a Square Pulse

$$F(\mu) = \int_{-\frac{W}{2}}^{\frac{W}{2}} A e^{-i2\pi\mu t} dt$$

$$= \frac{-A}{i2\pi\mu} \left[e^{-i2\pi\mu t} \right]_{-W/2}^{W/2}$$

$$= \frac{-A}{i2\pi\mu} \left[e^{-i\pi\mu W} - e^{i\pi\mu W} \right]$$

$$= \frac{A}{i2\pi\mu} \left[e^{i\pi\mu W} - e^{-i\pi\mu W} \right]$$



Fourier Transform of a Square Pulse

$$F(\mu) = \int_{-\frac{W}{2}}^{\frac{W}{2}} A e^{-i2\pi\mu t} dt$$

$$= \frac{-A}{i2\pi\mu} \left[e^{-i2\pi\mu t} \right]_{-W/2}^{W/2}$$

$$= \frac{-A}{i2\pi\mu} \left[e^{-i\pi\mu W} - e^{i\pi\mu W} \right]$$

$$= \frac{A}{i2\pi\mu} \left[e^{i\pi\mu W} - e^{-i\pi\mu W} \right]$$

$$F(\mu) = AW \frac{\sin(\pi\mu W)}{(\pi\mu W)}$$



Fourier Transform of a Square Pulse

$$F(\mu) = \int_{-\frac{W}{2}}^{\frac{W}{2}} A e^{-i2\pi\mu t} dt$$

$$= \frac{-A}{i2\pi\mu} \left[e^{-i2\pi\mu t} \right]_{-W/2}^{W/2}$$

$$= \frac{-A}{i2\pi\mu} \left[e^{-i\pi\mu W} - e^{i\pi\mu W} \right]$$

$$= \frac{A}{i2\pi\mu} \left[e^{i\pi\mu W} - e^{-i\pi\mu W} \right]$$

$$F(\mu) = AW \frac{\sin(\pi\mu W)}{(\pi\mu W)}$$

$$e^{i\theta} = \cos(\theta) + i\sin(\theta)$$

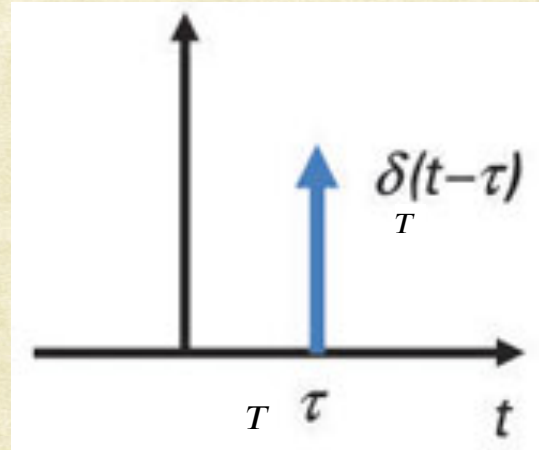


Questions?



Convolver with Shifted Impulse Function

Shifted impulse



$$\begin{aligned}\delta(t) &= 0, \text{ for } t \neq T \\ &= \infty, \text{ for } t = T\end{aligned}$$

$$\int_{-\infty}^{\infty} \delta(t - T) dt = 1$$

Sifting Property

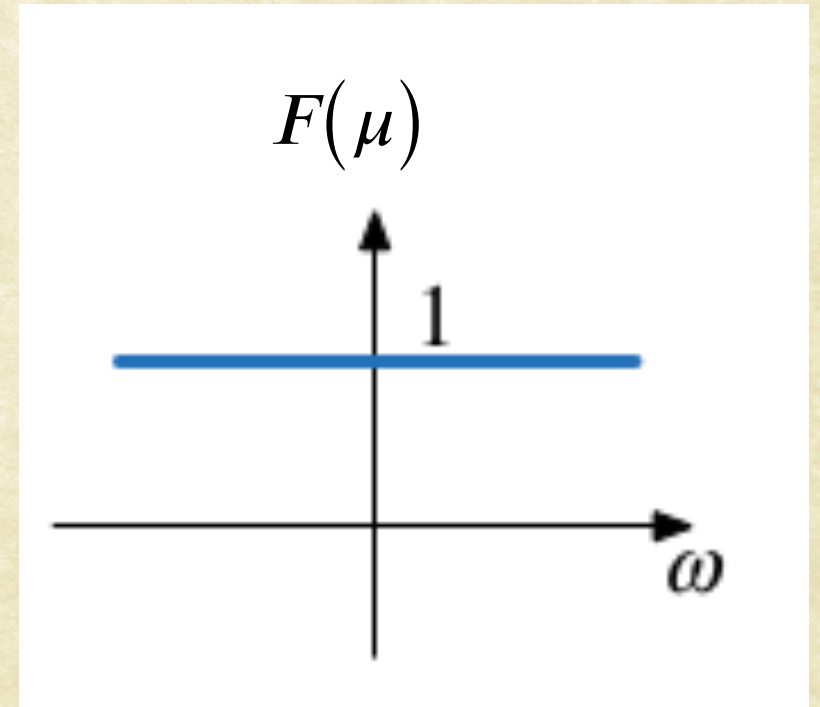
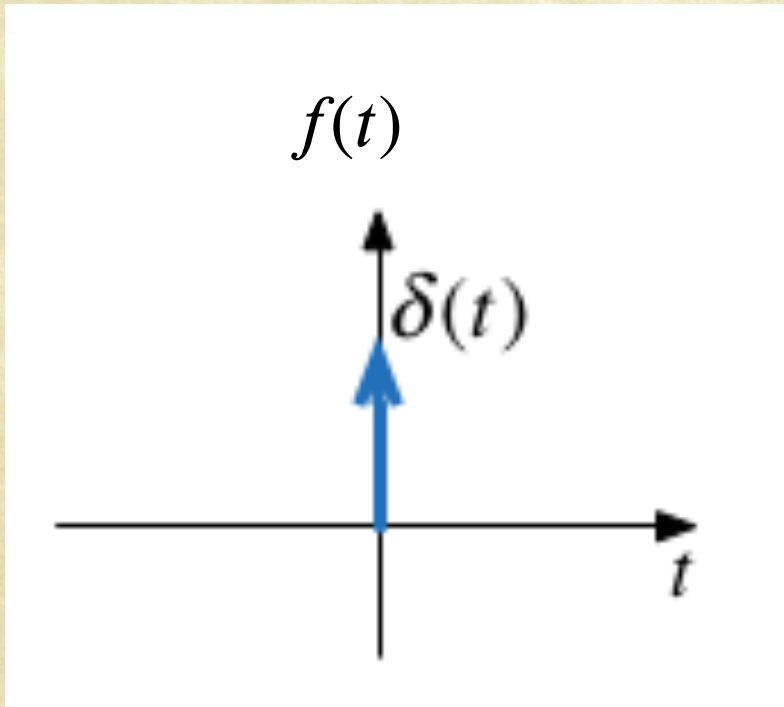
$$\begin{aligned}\int_a^b \delta(t - T) f(t) dt &= f(T), \quad a < T < b \\ &= 0 \text{ otherwise}\end{aligned}$$



FT of impulse function

$$f(t) = \delta(t)$$

$$F(\mu) = \int_{-\infty}^{\infty} \delta(t) e^{-i2\pi\mu t} dt$$

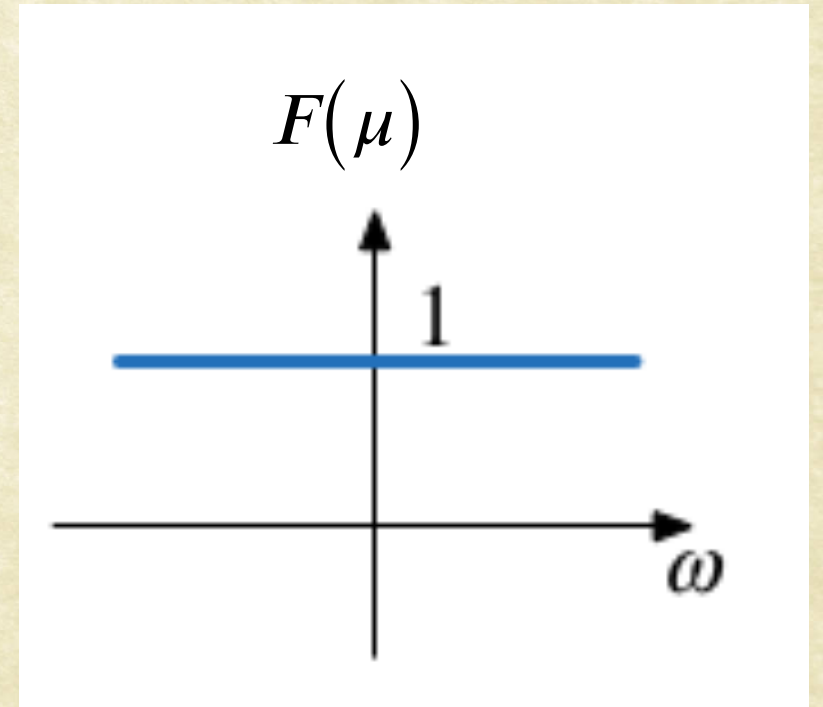
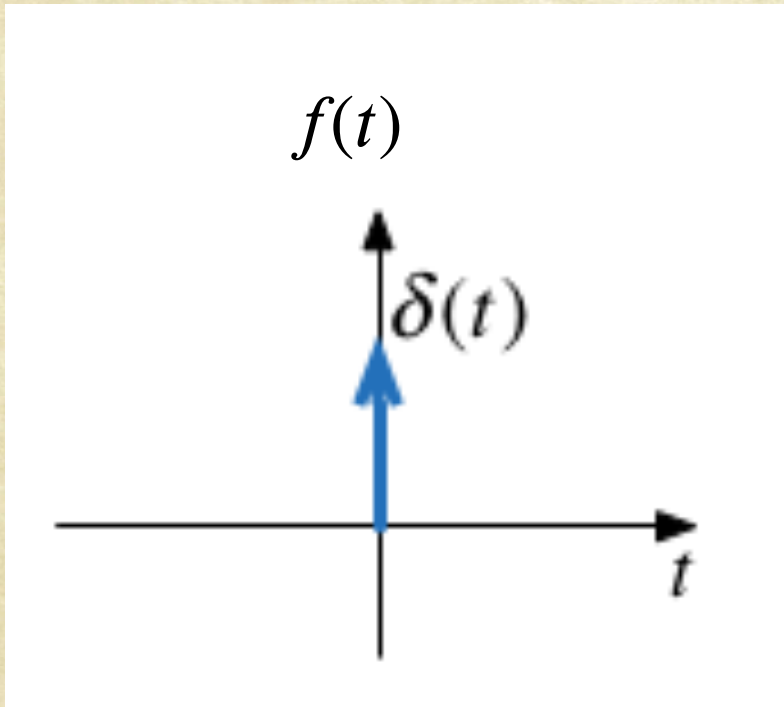




FT of impulse function

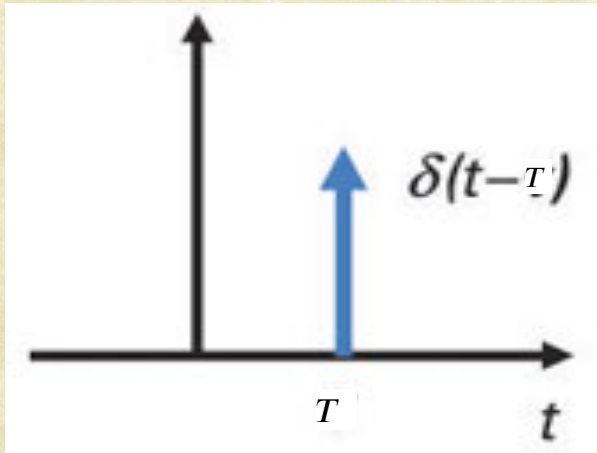
$$f(t) = \delta(t)$$

$$F(\mu) = \int_{-\infty}^{\infty} \delta(t) e^{-i2\pi\mu t} dt$$





FT of time-shifted impulse



$$\int_a^b \delta(t - t_0) f(t) dt = f(t_0), \quad a < t_0 < b$$

$$= 0, \quad \text{otherwise}$$

$$F(\mu) = \int_{-\infty}^{\infty} \delta(t - t_0) e^{-i2\pi\mu t} dt$$

$$= e^{-i2\pi\mu t_0}$$



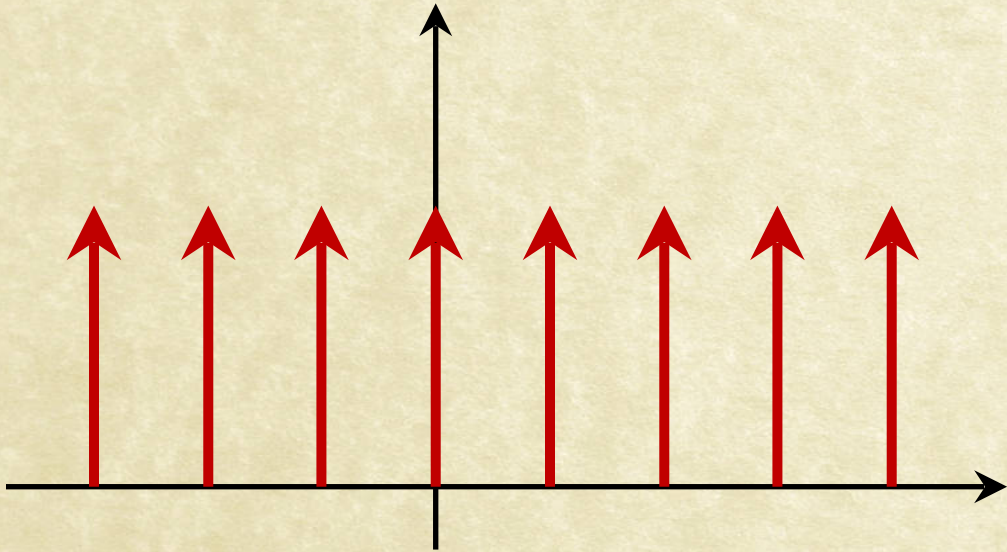
Questions?



FT of an Impulse Train

- We know that:

$$\mathcal{F}[\delta(t)] = e^{-i2\pi\mu t_0}$$

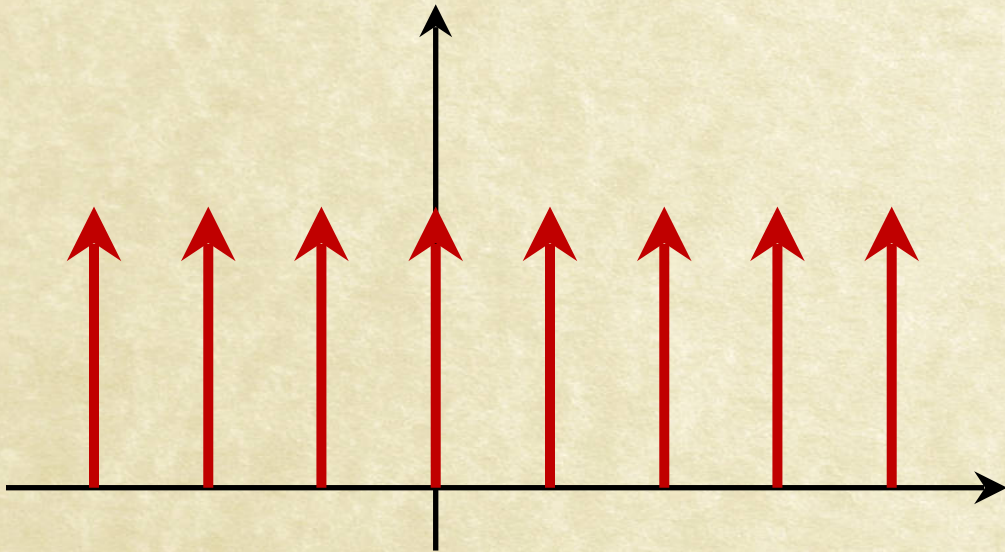




FT of an Impulse Train

- We know that:

$$\mathcal{F}[\delta(t)] = e^{-i2\pi\mu t_0}$$





FT of a Periodic Function

Fourier Series:

$$s_{\Delta T}(t) = \sum_{n=-\infty}^{\infty} c_n e^{i \frac{2\pi n}{\Delta T} t}$$

$$c_n = \frac{1}{\Delta T} \int_{-\Delta T/2}^{\Delta T/2} s_{\Delta T}(t) e^{-i \frac{2\pi n}{\Delta T} t} dt$$



The Impulse Train is:

$$s_{\Delta T}(t) = \sum_{n=-\infty}^{\infty} c_n e^{i \frac{2\pi n}{\Delta T} t}$$



The Impulse Train is:

$$s_{\Delta T}(t) = \sum_{n=-\infty}^{\infty} c_n e^{i \frac{2\pi n}{\Delta T} t}$$

$$c_n = \frac{1}{\Delta T} \int_{-\Delta T/2}^{\Delta T/2} \delta(t) e^{-i \frac{2\pi n}{\Delta T} t} dt$$



The Impulse Train is:

$$s_{\Delta T}(t) = \sum_{n=-\infty}^{\infty} c_n e^{i \frac{2\pi n}{\Delta T} t}$$

$$\begin{aligned} c_n &= \frac{1}{\Delta T} \int_{-\Delta T/2}^{\Delta T/2} \delta(t) e^{-i \frac{2\pi n}{\Delta T} t} dt \\ &= \frac{1}{\Delta T} e^0 = \frac{1}{\Delta T} \end{aligned}$$



The Impulse Train is:

$$s_{\Delta T}(t) = \sum_{n=-\infty}^{\infty} c_n e^{i \frac{2\pi n}{\Delta T} t}$$

$$= \frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} e^{i \frac{2\pi n}{\Delta T} t}$$

$$\begin{aligned} c_n &= \frac{1}{\Delta T} \int_{-\Delta T/2}^{\Delta T/2} \delta(t) e^{-i \frac{2\pi n}{\Delta T} t} dt \\ &= \frac{1}{\Delta T} e^0 = \frac{1}{\Delta T} \end{aligned}$$



FT of the Impulse Train

$$\mathcal{F}(s_{\Delta T}(t)) = \mathcal{F}\left(\frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} e^{i\frac{2\pi n}{\Delta T}t}\right)$$



FT of the Impulse Train

$$\mathcal{F}(s_{\Delta T}(t)) = \mathcal{F}\left(\frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} e^{i\frac{2\pi n}{\Delta T}t}\right)$$

$$= \frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} \mathcal{F}\left(e^{i\frac{2\pi n}{\Delta T}t}\right)$$



FT of the Impulse Train

$$\mathcal{F}(s_{\Delta T}(t)) = \mathcal{F}\left(\frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} e^{i\frac{2\pi n}{\Delta T}t}\right)$$

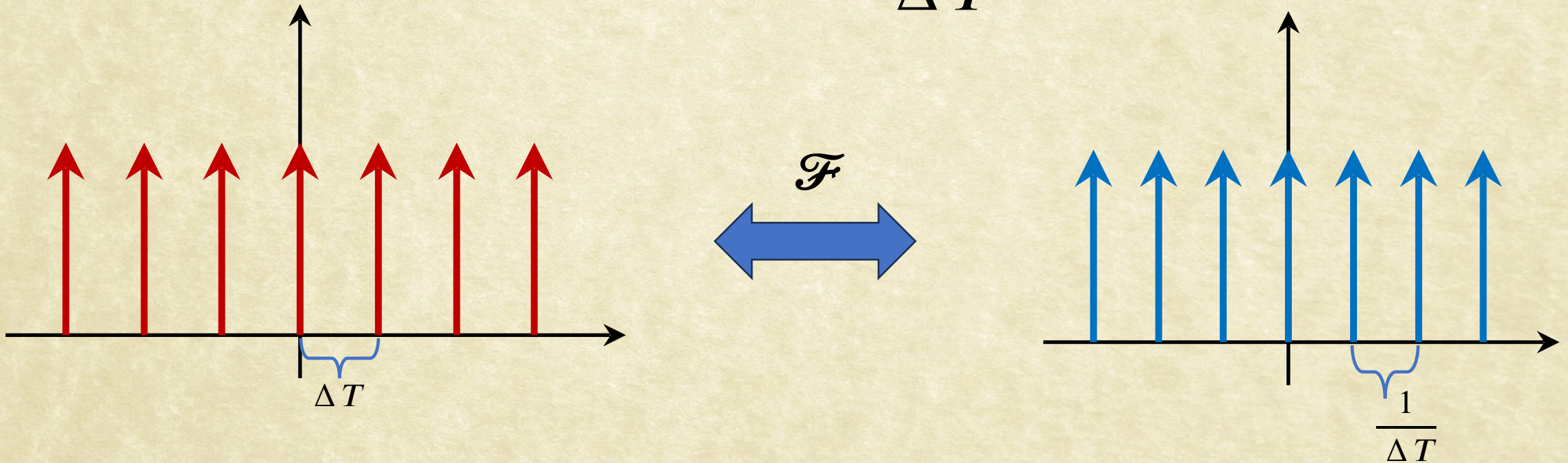
$$= \frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} \mathcal{F}\left(e^{i\frac{2\pi n}{\Delta T}t}\right)$$

$$= \frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} \delta\left(\mu - \frac{n}{\Delta T}\right)$$



Fourier Transform of an Impulse Train

- The FT of an Impulse train with period ΔT is an Impulse train with period $\frac{1}{\Delta T}$.





Questions?



FT of Convolution

$$(f \star h)(t) = \int_{-\infty}^{\infty} f(\tau)h(t - \tau)d\tau$$



FT of Convolution

$$(f \star h)(t) = \int_{-\infty}^{\infty} f(\tau)h(t - \tau)d\tau$$

$$\mathcal{F}\{(f \star h)(t)\} = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(\tau)h(t - \tau)d\tau \right] e^{-i2\pi\mu t} dt$$



FT of Convolution

$$(f \star h)(t) = \int_{-\infty}^{\infty} f(\tau)h(t - \tau)d\tau$$

$$\begin{aligned}\mathcal{F}\{(f \star h)(t)\} &= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(\tau)h(t - \tau)d\tau \right] e^{-i2\pi\mu t} dt \\ &= \int_{-\infty}^{\infty} f(\tau) \left[\int_{-\infty}^{\infty} h(t - \tau)e^{-i2\pi\mu t} dt \right] d\tau\end{aligned}$$



FT of Convolution

$$(f \star h)(t) = \int_{-\infty}^{\infty} f(\tau)h(t - \tau)d\tau$$

$$\mathcal{F}\{(f \star h)(t)\} = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(\tau)h(t - \tau)d\tau \right] e^{-i2\pi\mu t} dt$$

$$= \int_{-\infty}^{\infty} f(\tau) \left[\int_{-\infty}^{\infty} h(t - \tau)e^{-i2\pi\mu t} dt \right] d\tau$$

$$\mathcal{F}\{(f \star h)(t)\} = \int_{-\infty}^{\infty} f(\tau) [H(\mu)e^{-i2\pi\mu\tau}] d\tau$$



FT of Convolution



FT of Convolution

$$\mathcal{F}\{(f \star h)(t)\} = \int_{-\infty}^{\infty} f(\tau) [H(\mu) e^{-i2\pi\mu\tau}] d\tau$$



FT of Convolution

$$\begin{aligned}\mathcal{F}\{(f \star h)(t)\} &= \int_{-\infty}^{\infty} f(\tau) [H(\mu) e^{-i2\pi\mu\tau}] d\tau \\ &= H(\mu) \int_{-\infty}^{\infty} f(\tau) e^{-i2\pi\mu\tau} d\tau\end{aligned}$$



FT of Convolution

$$\begin{aligned}\mathcal{F}\{(f \star h)(t)\} &= \int_{-\infty}^{\infty} f(\tau) [H(\mu) e^{-i2\pi\mu\tau}] d\tau \\ &= H(\mu) \int_{-\infty}^{\infty} f(\tau) e^{-i2\pi\mu\tau} d\tau \\ &= H(\mu) F(\mu) = H \cdot F(\mu)\end{aligned}$$



Questions?